

NLP Case Study 1

MECE TABLE

Name	Task
Swetha Tanikonda	1 + merging
Gaurav Singh	2
Comfort Iroha Onuoha	3
Fatemi Sadikbhai Lokhandwala	3
Ogechi Tina Ukanwune	2
Sherin Stanley	4(Preprocessing, Feature engineering, Count Vectorisation)
Rahul Batherikal	4(SVM, lime, Cross Validation, next step)
Saraka Sahaswin Thanaweera Thanaweera Achchige Don	5

All the separate files that each individual worked on can be find in their respective branch

Prompt used:

Q3(Fatemi):

1. File reading, data filtering, prepossessing, naive bayes model evaluation, feature imbalance using smote and class weight
2. Feature engineering, hyperparameter tuning, feature importance and code improvement.

References:

1. <https://chatgpt.com/share/6795b0ad-b418-8010-85d9-81b8c3131f90>
2. <https://chatgpt.com/share/6795b095-b288-8010-a827-3cbf5a0e8caf>
3. <https://chatgpt.com/share/6795b0dc-554c-8010-b4b4-dfa5077d0f10>

Q5(Saraka) -

- First prompt - “Reading a json and convert to df and randomly pick 10000 lines python”
- Last prompt - “generate code to implement OVR AUC”

Q2 Gaurav

first: give e code to perform sentiment analysis on Yelp reviews using an SVM classifier. Start by loading 26,000 entries from the dataset, mapping star ratings to three sentiment labels: 'Positive' for 4-5 stars, 'Neutral' for 3 stars, and 'Negative' for 1-2 stars.

last: find the top 10 influential features per class from the SVM model and suggest improvements such as handling class imbalance or optimizing feature scaling.

Q1 swetha

- First: write the code to load the dataset that is in json format and take sample from it and then preprocess the text by converting it to lowercase, removing non-alphabetic characters, tokenizing, lemmatizing words, and eliminating stopwords
- Last: do multiclass roc curves and perform 5 fold cross validation for accuracy and generate LIME to explain and interpret influential words.

Q3(Tina)

- Extract sample file from the yelp dataset, text preprocessing and (8) negation handling, feature importance.

Q4.(Sherin) **SVM model using Count Vectors for Unigram+Bigram**

1. step by step procedure to Build the following models to predict whether a given review is Positive (4 or 5 stars), Negative (2 or less stars) or Neutral on the Yelp Dataset <https://www.yelp.com/dataset/download>SVM model using Count Vectors for Unigram+Bigram (ChatGpt)
2. numerical star ratings into sentiment categories: 'Positive' for ratings 4 or 5, 'Negative' for ratings 2 or below, and 'Neutral' for a rating of 3 (ChatGpt)
3. python code to Remove URLs, mentions, and digits. (ChatGpt)

[chatgpt](#)

Q4(Rahul) **SVM model using Count Vectors for Unigram+Bigram**

- SVM model using Count Vectors for Unigram+Bigram(chatGpt)
- Hyperparameter Tuning: Experiment with kernel functions (e.g., polynomial or sigmoid). Try a finer grid search on C, gamma, and other hyperparameters.(chatGpt)

- improve model results further by experimenting with polynomial and sigmoid kernel functions.(chatGpt)
- Evaluate the using metrics like accuracy, precision, recall, and F1-score(chatGpt)

Chatgpt : <https://chatgpt.com/share/6797d037-9954-8003-96e9-991854a02a12>

Result table

1. Naive Bayes model using Count Vectors for Unigram(swetha)

	Task	Status	Results	Individual Responsible
0	Config of the model trained	completed	CountVectorizer(ngram_range=(1,1)) + MultinomialNB(alpha=1.0) with MinMaxScaler (numeric) and OneHotEncoder (categorical).	Swetha
1	Train Time = ?	completed	0.80 seconds	Swetha
2	Training Data Check	completed	Sample Size: 20,800	Swetha
3	Confusion Matrix Built?	completed	yes	Swetha
4	F1 Score for Positive = ?	completed	Training: 0.93	Swetha
5	F1 Score for Negative = ?	completed	Training: 0.79	Swetha
6	AUC plotted?	completed	Yes	Swetha
7	AUC = ?	completed	AUC Values: Negative (0.92), Neutral (0.77), Positive (0.91).	Swetha
8	Accuracy computed?	completed	yes	Swetha
9	Accuracy = ?	completed	Training: 87%	Swetha
10	Feature Engg	NaN	NaN	NaN
11	Categorical/Numerical Features Added?	completed	Yes	Swetha
12	Name 2 features added?	completed	Numeric: useful, funny, cool\nCategorical: month	Swetha
13	Cross Validation	completed	5-fold CV Accuracy: 81% (± 0.00)	Swetha
14	Interpretability Implemented? Local or Global? 2 Interesting findings?	completed	Local (LIME): Used to explain individual predictions (e.g., text features).	Swetha
15	Testing Data Check	completed	Sample Size: 5,200	Swetha
16	Confusion Matrix Built?	completed	yes	Swetha
17	F1 Score for Positive = ?	completed	Test: 0.90	Swetha
18	F1 Score for Negative = ?	completed	Test: 0.72	Swetha
19	Accuracy computed?	completed	Yes	Swetha
20	Accuracy = ?	completed	Test: 81%	Swetha
21	Next Steps	NaN	NaN	NaN
22	List out 2-3 possible next steps for the this model	completed	1. Address class imbalance (e.g., Neutral class has low recall).\n2. Try advanced text embeddings (TF-IDF, BERT).\n3. Hyperparameter tuning for alpha in Naive Bayes.	Swetha

2. SVM model using TF-IDF Vectors with Unigram+Bigram

Gaurav

Task	Status	Results	Individual Responsible
Config of the model trained	done	SVM Model: kernel='linear', C=10.0, class_weight='balanced' TF-IDF: ngram_range=(1,2), max_features=10,000	Gaurav Singh
Train Time = ?	done	666.09 seconds	Gaurav Singh
Training Data Check	done	Sample Size: 20,800 Class Distribution: Positive (14,559), Negative (3,878), Neutral (2,363)	Gaurav Singh
Confusion Matrix Built?	done	Training Confusion Matrix: [3877, 0, 1] (Negative), [0, 2363, 0] (Neutral), [45, 133, 14381] (Positive)	Gaurav Singh
F1 Score for Positive = ?	done	Training: 0.99	Gaurav Singh
F1 Score for Negative = ?	done	Training: 0.99	Gaurav Singh
AUC plotted?	done	Yes (Multi-Class ROC curves for Test Set)	Gaurav Singh
AUC = ?	done	Negative: 0.95, Neutral: 0.80, Positive: 0.94	Gaurav Singh
Accuracy computed?	done	Yes	Gaurav Singh
Accuracy = ?	done	Training: 99%	Gaurav Singh
Feature Engg	done	TF-IDF with Unigrams+Bigrams (max_features=10,000)	Gaurav Singh
Categorical/Numerical Features	done	No (Only text features via TF-IDF)	Gaurav Singh
Name 2 features added?	done	Examples: "worst", "two stars" (Negative class); "mediocre", "bland" (Neutral class)	Gaurav Singh
Cross Validation	done	CV: F1-Macro = 0.66 (± 0.01)	Gaurav Singh
Interpretability Implemented? Local or Global? 2 Interesting findings?	done	Local; Findings: 1. Neutral class dominated by words like "mediocre" and "bland". 2. Positive class includes counterintuitive terms like "ok" and "unfortunately"	Gaurav Singh
Testing Data Check	done	Test Set Size: 5,200 samples	Gaurav Singh
Confusion Matrix Built?	done	Test Confusion Matrix: [738, 139, 92] (Negative), [161, 219, 211] (Neutral), [143, 252, 3245] (Positive)	Gaurav Singh
F1 Score for Positive = ?	done	Test: 0.90	Gaurav Singh
F1 Score for Negative = ?	done	Test: 0.73	Gaurav Singh
AUC plotted?	done	Yes (Multi-Class ROC curves for Test Set)	Gaurav Singh
AUC = ?	done	Negative: 0.95, Neutral: 0.80, Positive: 0.94, Macro-Average: 0.89	Gaurav Singh
Accuracy computed?	done	yes	Gaurav Singh
Accuracy = ?	done	Test: 81%	Gaurav Singh
Next Steps			
List out 2-3 possible next steps for the this model	done	1. Hyperparameter tuning (e.g., try kernel='rbf' as per GridSearchCV). 2. Address class imbalance (e.g., oversampling). 3. Feature scaling to resolve convergence warnings.	Gaurav Singh

Tina

Category	Details
SVM model using TF-IDF vectors and Unigram+Bigram	
Type of Smoothing Used?	SMOTE - OVER SAMPLING
Config of model trained?	YES
Train Time = ?	21.7638 seconds
Training Data Check	YES
Confusion Matrix Built?	YES
F1 Score for Positive = ?	1.00
F1 Score for Negative = ?	0.99
AUC plotted?	NO
AUC = ?	
Accuracy computed?	YES
Accuracy = ?	0.99
Feature Engg	review_length and polarity
Feature Weights Added?	YES
2 Features with the Highest Weights?	(trivia, 1.834247),(customer, 1.749706)
Cross Validation	YES
Type of Cross Validation performed?	K-FOLD
Findings of Cross Validation?	The mean cross-validation score is 94.72%, indicating strong overall performance
Next Steps	Testing Data Check
Testing Data Check	YES
Confusion Matrix Built?	YES
F1 Score for Positive = ?	0.90
F1 Score for Negative = ?	0.72
AUC plotted?	NO
AUC = ?	
Accuracy computed?	YES
Accuracy = ?	0.80
Next Steps	YES
List out 2-3 possible next steps for Naive Bayes Models	1. Hyperparameter Tuning: Use GridSearchCV or RandomizedSearchCV to optimize model hyperparameters. 2. Use Deep Learning: If performance plateaus, moving to deep learning techniques like BERT for text classification could yield significant improvements.

3. Naive Bayes model using TF-IDF Vectors for Unigram+Bigram

Fatemi

Que 3 - Naive Bayes model using TF-IDF Vectors for Unigram+Bigram

		Training	Testing
Evaluate the Naive Bayes Model	Train Time	0.14 seconds	-
	F1 Score for Positive	0.90	0.90
	F1 Score for Neutral	0.15	0.09
	F1 Score for Negative	0.76	0.73
	AUC	0.9257	0.9091
	Accuracy	0.8270	0.8180
SMOTE + Class Weights on Naive Bayes Model	Train Time	0.32 seconds	-
	F1 Score for Positive	0.83	0.86
	F1 Score for Neutral	0.73	0.43
	F1 Score for Negative	0.80	0.70
	AUC	0.9203	0.8913
	Accuracy	0.7851	0.7546
Hyperparameter tuning	Train Time	3.66 seconds	-
	F1 Score for Positive	0.91	0.90
	F1 Score for Neutral	0.22	0.14
	F1 Score for Negative	0.76	0.73
	AUC	0.9294	0.9078
	Accuracy	0.8342	0.8212

Feature Engg - Name 2 features added (Categorical/Numerical Features)

- sentiment (Categorical - derived from mapping stars to Positive, Neutral, Negative).
- cleaned_text (Categorical - preprocessed version of text with tokenization, stopword removal, and lemmatization).

Interpretability Implemented

- Global: The analysis identifies key features influencing the model's predictions across all reviews.

2 Interesting Findings

- Words like "food", "service", and "place" influence all sentiments but carry different contexts (e.g., "poor service" vs. "excellent service").
- "Great" is highly indicative of Positive sentiment, showing its strong association with positive experiences.

Type of Cross Validation performed: 5-fold cross-validation (via cv=5 in GridSearchCV).

Findings of Cross Validation:

The best hyperparameter ($\alpha = 0.01$) was selected based on the 5-fold cross-validation, optimizing model accuracy. The cross-validation helps prevent overfitting by ensuring the model is evaluated on different subsets of the data.

Comfort

Naive Bayes Model with One-Hot vectors and Unigram+Bigram	
Type of Smoothing Used?	OVER SAMPLING
Config of model trained?	YES
Train Time = ?	0.3944 seconds
Training Data Check	YES
Confusion Matrix Built?	YES
F1 Score for Positive = ?	0.910
F1 Score for Negative = ?	0.910
AUC plotted?	YES
AUC = ?	0.939
Accuracy computed?	YES
Accuracy = ?	0.905
Feature Engg	YES
Feature Weightages Added?	YES
2 Features with the Highest Weights?	(3.719248949015009, 'wow ed'), (3.631047779039765, 'pimento cheese')
Cross Validation	YES
Type of Cross Validation performed?	K-FOLD
Testing Data Check	YES
Confusion Matrix Built?	YES
F1 Score for Positive = ?	0.816
F1 Score for Negative = ?	0.816
AUC plotted?	YES
AUC = ?	0.854
Accuracy computed?	YES
Accuracy = ?	0.800

4. SVM model using Count Vectors for Unigram+Bigram

Sherin

1	Train Time = ?	done	258.1799736022949 seconds	Sherin Stanley
2	Confusion Matrix Built?	done	Training Confusion Matrix:\n[[2603 0 6]\n [3 1516 50]\n [0 4 9818]]	Sherin Stanley
3	F1 Score for Positive = ?	done	Training: 0.99	Sherin Stanley
4	F1 Score for Negative = ?	done	Training: 0.99	Sherin Stanley
5	Accuracy computed?	done	Yes	Sherin Stanley
6	Accuracy = ?	done	Training: 99%	Sherin Stanley
7	Feature Engg	done	CountVectorizer(ngram_range=(1, 2), stop_words='english', max_features=5000)	Sherin Stanley
8	Categorical/Numerical Features Added?	done	Yes	Sherin Stanley
9	Name 2 features added?	done	Examples: "Stars" to "Sentiments", "Reviews" to "cleaned text"	Sherin Stanley

Rahul

10	Cross Validation	done	Fitting 5 folds for each of 24 candidates, totalling 120 fits\nBest Parameters: {'C': 0.1, 'degree': 3, 'gamma': 'scale', 'kernel': 'linear'}\nClassification Report:\n precision recall f1-score support\n\n Negative 0.68 0.67 0.67 1119\n\n Neutral 0.35 0.24 0.28 672\n\n Positive 0.87 0.92 0.89 4209\n\n\n accuracy 0.79 6000\n\n macro avg 0.63 0.61 0.62 6000\n\nweighted avg 0.77 0.79 0.78 6000	Rahul Batherikal
11	Interpretability Implemented? Local or Global? 2 Interesting findings?	done	Local: LIME explains individual predictions by approximating the model's behavior for one specific instance. This means it shows why the model made a certain prediction for one sample.\n	Rahul Batherikal
12	Confusion Matrix Built?	done	\nConfusion Matrix (Test):\n[[800 173 146]\n [183 235 254]\n [220 313 367]]	Rahul Batherikal
13	F1 Score for Positive = ?	done	Test:0.788800720828644	Rahul Batherikal
14	F1 Score for Negative = ?	done	Test: 0.7851666666666667	Rahul Batherikal
15	AUC plotted?	done	Yes	Rahul Batherikal
16	AUC = ?	done	AUC: 0.8343655872341177	Rahul Batherikal
17	Accuracy computed?	done	yes	Rahul Batherikal
18	Accuracy = ?	done	Test: 0.7851666666666667	Rahul Batherikal
19	Next Steps	NaN	NaN	NaN
20	List out 2-3 possible next steps for the this model	done	Evaluate performance after lemmatization and compare it to the previous results. Experiment with different models like Random Forest or XGBoost to see if they outperform the SVM.	Rahul Batherikal

5. Naive Bayes model using One-Hot Vectors with Unigram+Bigram

Saraka

AT

[2]:		Category	Metric/Detail	Details
0	Model Configuration		Model Type	MultinomialNB
1	Training Data Check		Train Time	0.0326 seconds
2	Training Data Check		Confusion Matrix Built?	Yes
3	Training Data Check		F1 Score for Positive	0.88
4	Training Data Check		F1 Score for Negative	0.96
5	Training Data Check		AUC plotted?	Yes
6	Feature Engineering	Categorical/Numerical Features Added?		3
7	Feature Engineering	Name 2 Features Added?	review_length, exclamation_count	
8	Cross Validation	Cross Validation Implemented?		Yes
9	Testing Data Check		Confusion Matrix Built?	Yes
10	Testing Data Check		F1 Score for Positive	0.75
11	Testing Data Check		F1 Score for Negative	0.9
12	Testing Data Check		AUC plotted?	Yes
13	Testing Data Check		Accuracy	0.8205
14	Next Steps	2 Interesting Findings	Incorporate Advanced Feature Engineering	
15	Next Steps		Try Alternative Models with Hyperparameter Tuning	